An Introduction to Universal Artificial Intelligence

Chapter 2, Section 2.3: Statistical Inference and Estimation

Based on Hutter, Catt & Quarel

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Outline

Why Statistical Inference? I

The prediction problem. Throughout this book we want an agent to predict what comes next in a sequence, given what it has seen so far. To make any progress, we first assume we know *something* about the process that generates the sequence, and we begin with the simplest possible assumption.

The i.i.d. Bernoulli assumption

A binary sequence $x_1, x_2, \ldots$ is generated *independently and identically distributed* (i.i.d.) by a Bernoulli distribution with an *unknown* parameter θ (theta) that must be inferred from past observations.

Plain English. "i.i.d." means every symbol in the sequence is produced by the same random mechanism, and knowing some symbols tells you nothing extra about the others. $\theta \in [0, 1]$ is the unknown probability that any given symbol equals 1.

Why Statistical Inference? II

This assumption is strong: it says nothing depends on anything before it, which is often unrealistic. Later chapters relax it:

- **Chapter 3** allows the distribution to depend on *any* past elements (general sequence prediction / induction).
- **Chapter 4** covers efficient predictors for sequences where the next symbol depends only on a *bounded* number of past elements (finite-memory / Markov models).

We will study the estimation problem from two angles:

- the **frequentist** perspective (this deck), which treats θ as a fixed but unknown constant and studies estimators as functions of random data;
- the **Bayesian** perspective (next chapter), which treats θ itself as a random variable with a prior distribution that gets updated by evidence.

Why Statistical Inference? III

Parametric models. We assume the population/process can be fully described by a probability model with parameters θ drawn from a set Θ (Theta), often $\Theta \subseteq \mathbb{R}^d$. Inference then reduces to *searching for the correct* $\theta \in \Theta$. For our running example, the Bernoulli model $\mathrm{Bern}(x;\theta)$ has $\Theta = [0,1] \subset \mathbb{R}$.
For now we only want a single numerical best guess for θ given data $x_{1:n}$ (meaning the observations $x_1, \ldots, x_n$). Later we will also want a measure of *confidence* in that guess, e.g. a statement like $\mathbb{P}(\theta \in [0.4, 0.6] \mid \mathcal{D})$ (the probability that θ lies in an interval, given data $\mathcal{D}$), which requires interval estimation.

Example 2.3.1: The Sunrise Problem I

Question. How likely is it that the sun will *not* rise tomorrow? Consider the naive frequentist estimate

$$\frac{\text{days the sun did not rise}}{\text{days the Earth (or humans) have experienced}}.$$

Plain English. This just counts failures over total observed days.
Since the sun has never failed to rise on any recorded day, this ratio gives probability 0. But probability 0 means "impossible" – and declaring the event impossible, just because it has never happened yet, seems absurd. Solving this puzzle motivates the *Laplace rule* and many other estimation methods used throughout the book.

Example 2.3.1: The Sunrise Problem II

Formal setup.

1. Sample space $\Omega = \{\text{not rise}, \text{rise}\}^n$. Define i.i.d. $\{0, 1\}$-valued random variables $X_1, \ldots, X_n \sim \mathrm{Bernoulli}(\theta)$, with fixed but unknown θ. The variable X_t returns 1 iff the sun rose on day t: $\mathbb{P}(X_i = 1) = \theta$ and $\mathbb{P}(X_i = 0) = 1 - \theta$.
2. We observe values $x_{1:n}$ from the n i.i.d. random variables $X_1, \ldots, X_n$.
3. We wish to *estimate* θ using an estimator $t(x_{1:n})$ – a function of the observed data. This value is the outcome of the random variable $T_n(X_1, \ldots, X_n)$.
4. Based on the estimate of θ, we predict x_{n+1}: will the sun rise on the next day or not?

Example 2.3.1: The Sunrise Problem III

Why model the sun as i.i.d. Bernoulli at all? Historically (the problem is 18th-century, due to Laplace), the physical mechanism behind sunrise was not understood. Imagine explaining this to someone who has never seen the sky, and only knows that each "day" something called the sun either "rises" or does not. With no other information, i.i.d. and Bernoulli are not unreasonable modeling choices.

A more modern treatment would use domain knowledge from astrophysics: the sun is expected to keep rising for roughly the next 5 billion years, until it swells up and engulfs the Earth (assuming nothing else happens first). We will return to this example once we have built up the tools (the Laplace/KT estimator) to resolve the "probability zero" absurdity.

The Likelihood Function I

Idea. Perhaps the most important and intuitively appealing estimation procedure is *maximum likelihood*: pick the parameter value that makes the data we actually observed as probable as possible.

Likelihood function

For a parameter θ based on a sample of n random variables $X_1, \ldots, X_n$, the *likelihood function* is the joint probability density of the sample, viewed as a function of θ:

$$L(\theta) \;=\; L(\theta; x_1, \ldots, x_n) \;:=\; p_{X_1,\ldots,X_n}(x_1, \ldots, x_n; \theta).$$

Plain English. $L(\theta)$ answers: "if the true parameter were θ, how likely would this exact data sample have been?" It is a function of θ with the data $x_1, \ldots, x_n$ held fixed.

The Likelihood Function II

If the X_i are i.i.d. with pdf $p_X(x;\theta)$, the likelihood factorizes into a product:

$$L(\theta) \;=\; \prod_{i=1}^{n} p_X(x_i;\theta).$$

Plain English. $\prod_{i=1}^{n}$ (product from $i = 1$ to n) just multiplies the individual densities $p_X(x_i;\theta)$ together, one per data point – valid because independence lets probabilities of separate events multiply.

$L(\theta)$ can be thought of as the joint conditional probability $p(x_1, \ldots, x_n \mid \theta)$. For this factorization to hold we need the X_i to be *conditionally* i.i.d. given θ:

$$\mathbb{P}(x_1, \ldots, x_n \mid \theta) \;=\; \prod_{i=1}^{n} p(x_i \mid \theta).$$

The Likelihood Function III

Maximum likelihood estimator (MLE)

The MLE of θ is the value that maximizes the likelihood:

$$\widehat{\theta}_{ML} := \arg\sup_{\theta \in \Theta} L(\theta; x_1, \ldots, x_n).$$

Plain English. $\arg\sup_{\theta\in\Theta}$ ("argument of the supremum") means: search over every allowed parameter value θ in Θ and return the one that makes $L(\theta)$ largest.
In practice we use the *log-likelihood* $\ln L(\theta)$, since $\ln$ is monotonically increasing (so maximizing L or $\ln L$ gives the same θ), and logs turn products into sums, which are far easier to differentiate:

$$\arg\sup_{\theta\in\Theta} L(\theta) = \arg\sup_{\theta\in\Theta} \ln L(\theta), \qquad \ln L(\theta) = \ln\Big(\prod_{i=1}^{n} p_X(x_i;\theta)\Big) = \sum_{i=1}^{n} \ln p_X(x_i;\theta).$$

The derivative $\nabla_\theta \ln L(\theta)$ (gradient with respect to θ) is easier to compute than $\nabla_\theta L(\theta)$ because derivatives are linear operators, and setting it to zero locates the maximizing θ.

Example 2.3.2: MLE of a Bernoulli Sequence I

Suppose $X_1, \ldots, X_n \sim \mathrm{Bern}(x;\theta)$ with unknown θ, where

$$\mathrm{Bern}(x;\theta) \;=\; \theta^x (1-\theta)^{1-x}.$$

Plain English. This single formula gives θ when $x = 1$ and $1 - \theta$ when $x = 0$ – exactly the two Bernoulli outcomes, written compactly using exponents.
The log-likelihood is

$$\ln L(\theta) \;=\; \sum_{i=1}^{n} \big[x_i \ln\theta + (1 - x_i)\ln(1-\theta)\big].$$

Example 2.3.2: MLE of a Bernoulli Sequence II

Differentiating with respect to θ:

$$\nabla_\theta \ln L(\theta) \;=\; \sum_{i=1}^{n} \left[\frac{x_i}{\theta} + \frac{x_i - 1}{1-\theta} \right].$$

Setting this equal to zero and solving for θ gives the MLE:

$$\boxed{\widehat{\theta}_{ML} \;=\; \frac{1}{n} \sum_{i=1}^{n} x_i}$$

Plain English. The MLE of a Bernoulli parameter is simply the *sample average* – the fraction of 1's observed. This matches intuition: since $\mathbb{E}[X] = \theta$ for $X \sim \mathrm{Bern}(\theta)$, averaging samples estimates the expectation, which equals θ.
We call estimators of this "proportion of 1's" form *frequency estimators*.

Example 2.3.2: MLE of a Bernoulli Sequence III

Back to the sunrise problem. The MLE gives $\widehat{\theta}_{ML} = 1$ whenever every observed day the sun rose (no zeros at all in the data). This implies the sun will rise tomorrow with *absolute certainty* – again the nonsensical, overconfident answer.
More formally: if $\mathbb{P}(A) = 1$, then for any event B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B) = 1.$$

Plain English. Once you assign probability 1 to an event, no future evidence B can ever change that probability – the posterior is "stuck". This is an undesirable property for an estimator: it should stay open to revision.

Reparametrization Equivariance of the MLE

We parameterized the Bernoulli model by θ, the probability of observing a 1. But we could equally have chosen a different parametrization – e.g. the probability of observing a 0, or the *square* of the probability of observing a 1. In general we can reparameterize via $\tau = \psi(\theta)$ (tau equals psi of theta) for any bijection ψ (psi). We would expect the MLE to be *invariant* under this change: $\widehat{\tau}_{ML} = \psi(\widehat{\theta}_{ML})$.

Proposition 2.3.3 (Reparametrization equivariance of the MLE)

Let $x_1, \ldots, x_n$ be i.i.d. samples with likelihood function $L(\theta; x_1, \ldots, x_n)$, and let $\widehat{\theta}_{ML}$ be the MLE of θ. For any bijective function τ, define the induced likelihood

$$\tilde{L}(\tau(\theta); x_1, \ldots, x_n) \;:=\; L(\theta; x_1, \ldots, x_n),$$

and let $\widehat{\tau}_{ML}$ maximize $\tilde{L}$. Then

$$\widehat{\tau}_{ML} \;=\; \tau(\widehat{\theta}_{ML}).$$

Plain English. It does not matter whether you first find the MLE of θ and then transform it, or directly maximize likelihood in the new parametrization τ – you land on the same answer. Relabeling parameters cannot change what the "most likely" explanation of the data is.

Motivation: Beyond a Single Good Property

The MLE is intuitively appealing, but it has problems – it depends wholly on the data present, which can lead to *overfitting* when little data is available (as the sunrise problem illustrated). This section studies whether the MLE (and estimators in general) behave well *asymptotically*, as the number of samples $n \to \infty$.

For a sequence of i.i.d. random variables $X_1, X_2, \ldots$ controlled by parameter θ, an *estimator* T_n is a measurable function of only the first n variables:

$$T_n : \mathcal{X}_1 \times \cdots \times \mathcal{X}_n \to \Theta, \qquad \text{i.e.} \quad T_n(X_1, \ldots, X_n) \in \Theta,$$

where $\mathcal{X}_i = X_i(\Omega)$ is the alphabet (set of possible outcomes) of X_i. Since T_n is a function of random variables, T_n is itself a random variable. We study its *asymptotic behavior* as $n \to \infty$.

Definition 2.3.4: Estimator Consistency

Definition 2.3.4 (Estimator consistency)

Given a sequence of i.i.d. random variables $(X_i)_{i=1}^{\infty}$ with distribution controlled by parameter $\theta \in \Theta$, an estimator T_n for θ is *consistent* if

$$T_n \xrightarrow{P} \theta.$$

Plain English. $\xrightarrow{P}$ means *convergence in probability*: as we gather more and more samples ($n \to \infty$), the estimator's value gets arbitrarily close to the true parameter θ with probability approaching 1.
Informally: more data $\Rightarrow$ better estimate, with (arbitrarily) high confidence.
Example. The estimator $T_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ for a Bernoulli θ is consistent (and in fact unbiased – see below), since

$$\mathbb{E}[T_n] = \theta \qquad \text{and} \qquad \mathbf{Var}[T_n] = \frac{\theta(1-\theta)}{n} \longrightarrow 0.$$

Plain English. As n grows, the spread (variance) of the estimator around its correct average value shrinks to zero, so it concentrates tightly on θ.
Consistency alone is a fairly *weak* property: an estimator can still be biased – systematically over- or under-estimating θ – even while converging in the limit.

Definition 2.3.5 (Estimator bias)

Given $(X_i)_{i=1}^{\infty}$ as above, the *bias* of an estimator T_n is

$$\mathrm{Bias}(T_n) \ := \ \mathbb{E}_\theta[T_n] - \theta.$$

If $\mathrm{Bias}(T_n) = 0$ for all $n \geq 1$ and all $\theta \in \Theta$, then T_n is called *unbiased*. Here $\mathbb{E}_\theta[T_n]$ means the expectation taken with respect to the pdf $p_X(\cdot;\theta)$.

Plain English. Bias measures how far the estimator's *average* value (over many hypothetical repetitions of the experiment) is from the true θ. Zero bias means it is correct on average, even though any single estimate may be off.

Definition 2.3.5: Estimator Bias, and Two Contrasting Examples II

Example 2.3.6 (Consistent but biased estimator)

Given $\{x_i\}_{i=1}^{\infty}$ i.i.d. from $\mathrm{Bern}(\theta)$, the estimator $T_n = \left(\frac{1}{n}\sum_i x_i\right) + \frac{1}{n}$ for θ is biased, since

$$\mathrm{Bias}(T_n) = \mathbb{E}\Big[\big(\tfrac{1}{n}\sum_i x_i\big) + \tfrac{1}{n}\Big] - \theta = \mathbb{E}\big[\tfrac{1}{n}\sum_i x_i\big] - \theta + \tfrac{1}{n} = \tfrac{1}{n}.$$

Plain English. T_n overestimates θ by exactly $\frac{1}{n}$ *on average* – but since $\frac{1}{n} \to 0$ as $n \to \infty$, $T_n \xrightarrow{P} \theta$ still holds, so T_n is *consistent* despite being biased for every finite n.

Example 2.3.7 (Unbiased but inconsistent estimator)

The estimator $T_n = X_1$ (i.e. always just output the first sample) is unbiased, since $\mathbb{E}[X_1] = \theta$, but it fails to be consistent: T_n is always exactly 0 or exactly 1 for every n, which never converges to θ for any $\theta \in (0, 1)$.

Plain English. $T_n = X_1$ is "correct on average" across many repeated experiments, but within any *single* run it never improves with more data – it just ignores all data except the first sample. Unbiasedness and consistency are genuinely different, independent properties: neither implies the other.

Definition 2.3.8 and Theorem 2.3.9: Mean Squared Error

Checking consistency and bias is a good start, but we would like something *quantitative*: how far off is a typical estimate, and how fast does it converge?

Definition 2.3.8 (Estimator mean squared error)

The *mean squared error* (MSE) of an estimator T_n is

$$\mathrm{MSE}_\theta[T_n] \;:=\; \mathbb{E}_\theta\left[(T_n - \theta)^2\right].$$

Plain English. MSE is the average squared distance between the estimator and the true value – penalizing both systematic bias and random spread (variance).

Theorem 2.3.9 (MSE of an estimator)

If T_n is an estimator, then

$$\mathrm{MSE}_\theta[T_n] \;=\; \mathbf{Var}_\theta[T_n] + \mathrm{Bias}(T_n)^2.$$

Plain English. Total error-squared splits cleanly into two additive parts: how *spread out* the estimator is ($\mathbf{Var}$), plus how *off-center* it is (Bias^2). For an unbiased estimator, MSE coincides exactly with the variance.

Proof of Theorem 2.3.9

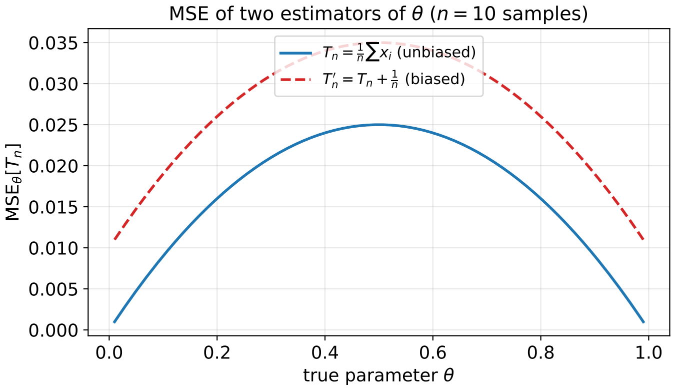
Proof. Let $t := \mathbb{E}_\theta[T_n]$. Then

$$\begin{aligned}
\mathrm{MSE}[T_n] &\equiv \mathbb{E}_\theta\left[(T_n - t + t - \theta)^2\right] \\
&= \mathbb{E}_\theta\left[(T_n - t)^2\right] + 2\big(\mathbb{E}_\theta[T_n] - t\big)(t - \theta) + (t - \theta)^2 \\
&= \mathbf{Var}_\theta[T_n] + 2\,\underbrace{\big(\mathbb{E}_\theta[T_n] - t\big)}_{=0}(t - \theta) + \mathrm{Bias}(T_n)^2 \\
&= \mathbf{Var}_\theta[T_n] + \mathrm{Bias}(T_n)^2. \qquad \blacksquare
\end{aligned}$$

Plain English. We add and subtract the mean t inside the square, expand, and note the cross term vanishes because $\mathbb{E}_\theta[T_n] - t = 0$ by definition of t. What remains is exactly variance plus bias-squared. We can now compare two estimators objectively by comparing their MSEs.

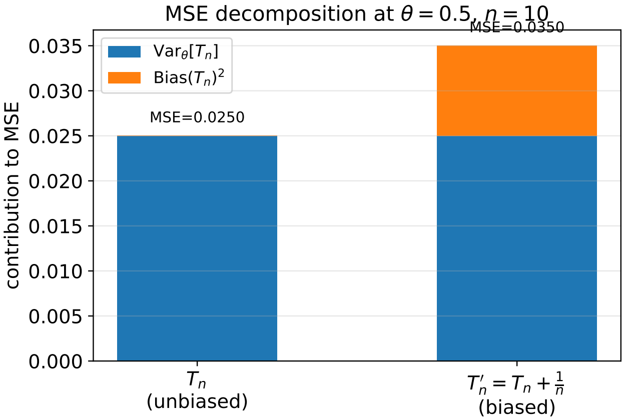
Visualizing the Bias–Variance Trade-off

Recall Example 2.3.6: $T_n = \frac{1}{n}\sum x_i$ (unbiased) versus $T'_n = T_n + \frac{1}{n}$ (biased by exactly $\frac{1}{n}$), both with $n = 10$ samples. Both share the same variance $\mathbf{Var}_\theta[T_n] = \theta(1-\theta)/n$, but T'_n carries an extra constant $\mathrm{Bias}^2 = 1/n^2$ term.



Reading the plot. The dashed (biased) curve sits slightly above the solid (unbiased) curve everywhere — a small, constant MSE penalty from the bias term that does not depend on θ. Neither

Bias–Variance–MSE Breakdown at $\theta = 0.5$



Reading the plot. Each bar stacks $\mathbf{Var}_\theta[T_n]$ (blue) below $\mathrm{Bias}(T_n)^2$ (orange); the bar's total height is $\mathrm{MSE}_\theta[T_n]$. Here both estimators have identical variance ($\theta(1-\theta)/n$), so all the difference in total MSE comes from the bias term of T'_n.

Definition 2.3.10: Estimator Domination

Now that we have a way to compare estimators quantitatively, we can ask: is there always a single best choice that beats every other estimator?

Definition 2.3.10 (Estimator domination)

An estimator T is said to *dominate* another estimator T' if, for all choices of parameter $\theta \in \Theta$,

$$\mathrm{MSE}_\theta[T] \ \leq \ \mathrm{MSE}_\theta[T'].$$

Plain English. T dominates T' if T is never worse, in expected squared error, no matter what the true (unknown) θ happens to be. This gives only a *partial ordering* – two estimators might each be better than the other for different ranges of θ, as in the plot on the previous slide, where the curves nearly overlap and neither strictly dominates.
This raises a natural question: is there a lower bound on how small the MSE of an unbiased estimator can possibly be? This motivates studying the *score* of a distribution, which underlies the Cramér–Rao Bound.

Definition 2.3.11: The Score

Definition 2.3.11 (Score)

Given a random variable $X \sim p_X(x;\theta)$ with distribution parameterized by θ, the *score* V of X is the random variable

$$V = \nabla_\theta \ln p_X(X;\theta).$$

For a family of random variables $X_1, \ldots, X_n \sim p_{X_1,\ldots,X_n}(x_1, \ldots, x_n; \theta)$, the *$n$-sample score* V_n is defined analogously using the joint distribution:

$$V_n = \nabla_\theta \ln p_{X_1,\ldots,X_n}(X_1, \ldots, X_n; \theta).$$

Plain English. The score is the gradient (slope) of the log-likelihood with respect to θ, evaluated at the (random) data itself. It measures *how sensitive* the log-likelihood is to small changes in θ – and recall that setting this very quantity to zero is exactly how we found the MLE.
One can show that the expected value of the score is zero, $\mathbb{E}_\theta[V] = 0$, and consequently

$$\mathbf{Var}_\theta[V] = \mathbb{E}_\theta[V^2].$$

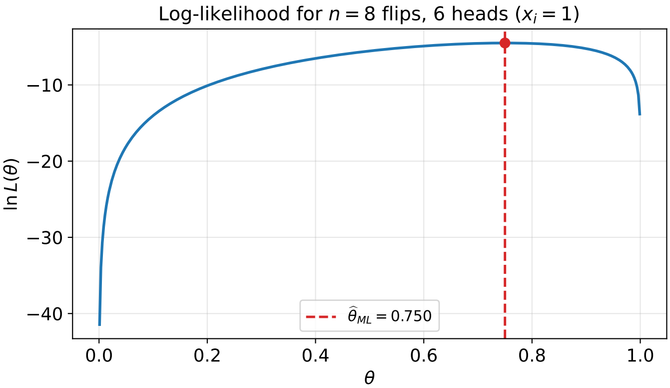
Plain English. Because the score averages to zero, its variance is simply the average of its square – this quantity is called the *Fisher information*, and it will be used (beyond what is covered in this deck) to bound how good *any* unbiased estimator can be.

Python: Maximum Likelihood Estimation for a Bernoulli Sample

We compute $\widehat{\theta}_{ML} = \frac{1}{n}\sum_i x_i$ directly from a small concrete sample, and plot the log-likelihood to confirm the MLE sits exactly at its peak.

```python
import numpy as np

# A concrete coin-flip sample: 1 = heads (sun rose), 0 = tails
x = np.array([1, 0, 1, 1, 0, 1, 1, 1])
n, s = len(x), x.sum()              # n = 8 flips, s = 6 heads

theta_hat_ml = s / n
print(f"MLE estimate theta_hat_ML = {theta_hat_ml:.3f}")
# -> MLE estimate theta_hat_ML = 0.750

# Log-likelihood as a function of theta, for sanity-checking the MLE
def log_likelihood(theta, s, n):
    return s * np.log(theta) + (n - s) * np.log(1 - theta)

thetas = np.linspace(0.01, 0.99, 9901)
theta_argmax = thetas[np.argmax(log_likelihood(thetas, s, n))]
print(f"Numerically maximized theta = {theta_argmax:.3f}")
# -> Numerically maximized theta = 0.750  (matches closed form s/n)
```

The Log-Likelihood Curve, and its Maximum



Reading the plot. With $n = 8$ flips and $s = 6$ heads, $\ln L(\theta)$ is concave and peaks exactly at $\widehat{\theta}_{ML} = 6/8 = 0.75$ (red dashed line and dot) – confirming the closed-form MLE derived by setting $\nabla_\theta \ln L(\theta) = 0$.

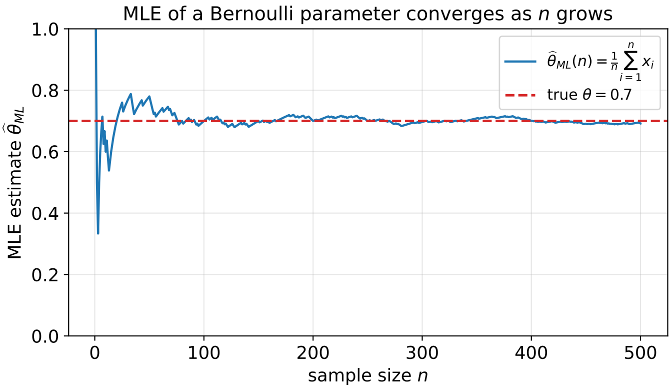
Python: Consistency – MLE Converges as n Grows

We simulate a long Bernoulli($\theta = 0.7$) sequence and track the running MLE estimate $\widehat{\theta}_{ML}(n)$ to see convergence in probability in action.

```
import numpy as np
rng = np.random.default_rng(42)

theta_true = 0.7
n_max = 500
x = (rng.random(n_max) < theta_true).astype(float)

running_mean = np.cumsum(x) / np.arange(1, n_max + 1)
print("Estimate after n=10  flips:", round(running_mean[9], 3))
print("Estimate after n=100 flips:", round(running_mean[99], 3))
print("Estimate after n=500 flips:", round(running_mean[-1], 3))
# -> Estimate after n=10  flips: 0.800   (noisy -- far from 0.7)
# -> Estimate after n=100 flips: 0.700   (much closer)
# -> Estimate after n=500 flips: 0.702   (very close to true theta)
```

As n grows, $\widehat{\theta}_{ML}(n)$ settles down near the true $\theta = 0.7$, illustrating $T_n \xrightarrow{P} \theta$ (Definition 2.3.4).

Visualizing Consistency



Reading the plot. The blue curve is the running MLE estimate $\widehat{\theta}_{ML}(n)$ from the code on the previous slide; the red dashed line is the true $\theta = 0.7$. Early on (small n) the estimate is noisy and can be far off, but it settles down and converges to the true value as n grows – exactly the behavior Definition 2.3.4 formalizes.

Python: Bias and MSE via Monte Carlo Simulation

We empirically verify Theorem 2.3.9 ($\mathrm{MSE} = \mathbf{Var} + \mathrm{Bias}^2$) for the biased estimator $T_n' = T_n + \frac{1}{n}$ of Example 2.3.6, by simulating many independent experiments.

```python
import numpy as np
rng = np.random.default_rng(0)

theta_true, n, n_trials = 0.5, 10, 200_000

# Simulate n_trials independent experiments, each with n Bernoulli flips
X = (rng.random((n_trials, n)) < theta_true).astype(float)
T  = X.mean(axis=1)            # unbiased estimator T_n
Tp = T + 1.0 / n               # biased estimator T_n' = T_n + 1/n

bias_T,  bias_Tp  = T.mean()-theta_true, Tp.mean()-theta_true
mse_T,   mse_Tp   = ((T-theta_true)**2).mean(), ((Tp-theta_true)**2).mean()
var_T,   var_Tp   = T.var(),  Tp.var()

print(f"T_n:  bias={bias_T:+.4f}  var={var_T:.5f}  MSE={mse_T:.5f}  (Var+Bias^2={var_T+bias_T**2:.5f})")
print(f"T_n': bias={bias_Tp:+.4f}  var={var_Tp:.5f}  MSE={mse_Tp:.5f}  (Var+Bias^2={var_Tp+bias_Tp**2:.5f})")
# -> T_n:  bias~+0.0000  var~0.02500  MSE~0.02500  (Var+Bias^2 ~ 0.02500)
# -> T_n': bias~+0.1000  var~0.02500  MSE~0.03500  (Var+Bias^2 ~ 0.03500)
```

The empirical MSE matches $\mathbf{Var} + \mathrm{Bias}^2$ to simulation precision for both estimators, confirming Theorem 2.3.9 numerically.

Summary: Section 2.3

- **Sunrise problem (2.3.1):** naive frequency estimation assigns probability 0 to never-observed events – an absurd, overconfident conclusion that motivates better estimators.
- **Maximum likelihood (2.3.2):** $\widehat{\theta}_{ML}$ maximizes $L(\theta) = \prod_i p_X(x_i; \theta)$; for Bernoulli data, $\widehat{\theta}_{ML} = \frac{1}{n} \sum_i x_i$ (the sample mean), but it can be overconfident with little data.
- **Reparametrization equivariance (2.3.3):** the MLE commutes with any bijective relabeling of the parameter space.
- **Consistency and bias (2.3.4):** $T_n \xrightarrow{P} \theta$ (consistency) and $\mathbb{E}_\theta[T_n] = \theta$ (unbiasedness) are *independent* properties – neither implies the other.
- **MSE (Def. 2.3.8–2.3.10):** $\mathrm{MSE}_\theta[T_n] = \mathbf{Var}_\theta[T_n] + \mathrm{Bias}(T_n)^2$ lets us compare estimators quantitatively via *domination*.
- **Score (2.3.11):** $V = \nabla_\theta \ln p_X(X; \theta)$ has mean zero and variance $\mathbb{E}_\theta[V^2]$ – the Fisher information, setting the stage for a lower bound on estimator MSE (Cramér–Rao Bound).

Next: the Bayesian approach to estimation (Section 2.4), which treats θ itself as random and updates a full posterior distribution over it as evidence arrives.